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Studierichting: Informatica
Oefeningenreeks 2E nr. 26.8

$$\log_3(10 - 3^x) = \log_3(81) + \log_9\left(\frac{1}{81}\right) - x$$

$$\Leftrightarrow \log_3(10 - 3^x) = 4 + \log_9(1) - \log_9(81) - x$$

$$\Leftrightarrow \log_3(10 - 3^x) = 4 + 0 - 2 - x$$

$$\Leftrightarrow \log_3(10 - 3^x) = 2 - x$$

$$\Leftrightarrow 3^{2-x} = 10 - 3^x$$

$$\Leftrightarrow \frac{9}{3^x} = 10 - 3^x \text{ stel} 3^x = t$$

$$\Leftrightarrow \frac{t^2 - 10t + 9}{t} = 0$$

$$\Delta = 100 - 36 = 64$$

$$t_{1,2} = \frac{10 \pm 8}{2}$$

$$t_1 = 9 \; t_2 = 1$$

$$3^x = 9 \Leftrightarrow x = 2$$

$$3^x = 1 \Leftrightarrow x = 0$$

dus $x = 2 \vee x = 0$

References